

Lesson 5

Recap



Everything from Lessons 1 to 4.

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- 01 Data types & errors
 - 02 Showing data on the INCIPE Board screen
 - 03 Input, compare & decide
 - 04 Sensors & data cleaning
 - 05 Arrays & for loops
 - 06 Classwork: make a minigame
-

Six data types.

Type	Meaning	Example
Integer	Integer in Maths, can be positive, 0 or negative	1, 102
Float	Numbers that can have up to 7 decimal places	1.234, 2.345
Double	Numbers that can have up to 15 decimal places	1.234567890123
Char	Single character	A
String	Multiple characters	ABC, Hello
Boolean	A data type whose value is either true or false	true, false

Syntax, run-time & logic errors.

Syntax error

Written mistakes, eg a missing “;” or bracket.

Run-time error

The computer could not perform what you tell it to do, eg 1/0.

Logic error

The code runs, but the outcome is not what you expected.

LOGIC ERROR EXAMPLE

```
if (sensorValue > threshold) {  
    Print("true"); }  
else if (sensorValue < threshold) {  
    Print("false"); }
```

Nothing would be performed when sensorValue = threshold.

Checkpoint 1

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Which is the correct format for creating a variable?

READ THE CODE

```
type name = value;
```

```
int a = 10;
```

```
float t = 25.5;
```

```
String s = "Hi";
```

ANSWER

```
int a = 10;
```

WHY

Data type first, then the variable name, then = and a value, and end with “;”.

Getting input into the program.

Input	Data that goes into the program
Process	What the program does with the data
Output	The result shown to the user

WITHOUT A LIBRARY

```
while (Serial.available() == 0) {}  
int n = Serial.parseInt();  
float f = Serial.parseFloat();
```

WITH THE INCIPE LIBRARY

```
incipe.getTemperature();  
incipe.getLightIntensity();  
incipe.getDistance();
```

Three questions, three steps.

- 1** How to ask users to input two numbers?
How to store them?

```
Serial.println("Please type your first number.");  
while(Serial.available()==0){}  
int a = Serial.parseInt();
```

- 2** How to add the two numbers? Store it
to where?

```
int sum = a+b;
```

- 3** How to show the answer?

```
Serial.print("The answer is:");  
Serial.println(sum);
```

Compare, combine, decide.

COMPARE

Symbol	Meaning
<code>==</code>	Equals to
<code>!=</code>	Not equals to
<code><=</code>	Smaller than or equal to
<code>>=</code>	Larger than or equal to
<code><</code>	Smaller than
<code>></code>	Larger than

Remember: `=` assigns a value; `==` compares two values.

COMBINE

`&&` AND: both must be true

`||` OR: any one is true

`!` NOT

DECIDE

```
if (...) { } else if (...) { } else { }
```

What is the value of data?

READ THE CODE

```
int data = 8;
if (data < 5) {
    data = data + 5;
}
else if (data > 8) {
    data = data - 3;
}
else {
    data++;
}
```

ANSWER

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WHY

8 < 5 is false and 8 > 8 is false, so the else runs: data++ makes 8 into 9.

Which values make this true?

READ THE CODE

```
(x > y) && !(y ≥ z)
```

ANSWER

x=3, y=1, z=2

WHY

Both sides of && must be true: $x > y$, and $!(y \geq z)$ means $y < z$. Any $x > y$ with $y < z$ works.

Sensor readings need cleaning.

FUNCTIONS FOR SENSORS

```
incipe.getTemperature()  
incipe.getHumidity()  
incipe.getDistance()  
incipe.getLightIntensity()  
incipe.getPPM()  
incipe.getCOPPM()  
incipe.getAlcoholPPM()  
incipe.getFlameIntensity()
```

DATA CLEANING

Fixing or removing incorrect or incomplete raw data. Here, we focus on interpreting the raw data to make it more useful.

```
if AQvalue < xxxx  
    print Good  
else if AQvalue > xxxx and < xxxx  
    print Moderate  
else  
    print Poor
```

Create an array: type, name, size.

```
int data[6];
```

int array with a size of 6, without existing data

```
int value[5] = {2, 4, -8, 3, 2};
```

int array with a size of 5, with existing data

```
float number[2] = {2.8943, 10.23189};
```

You can use other data types too: float, size of 2

Write, read & clear by index.

Write

```
int data[6] = {1,2,3,4,5,6};  
data[1] = 10;
```

→ {1, 10, 3, 4, 5, 6}

data[1] is the second element. To change the first, use [0].

Read

```
int data[6] = {1,10,3,4,5,6};  
x = data[4];
```

→ x = 5

x gets a copy of the fifth element. The array does not change.

Clear

```
int data[6] = {1,10,3,4,5,6};  
data[5] = 0;
```

→ {1, 10, 3, 4, 5, 0}

We assume 0 means nothing, so this removes the sixth element.

What is in the array now?

READ THE CODE

```
int array[5] = {1, 3, 5, 7, 9};  
array[1] = array[2] + 3;  
array[4] = array[3] - 2;
```

ANSWER

{1, 8, 5, 7, 5}

WHY

$\text{array}[1] = 5 + 3 = 8$ and $\text{array}[4] = 7 - 2 = 5$. Index starts at 0.

Repeat a command with a fixed structure.

A for loop repeats some command. It is mainly used to control arrays.

```
for (int i = 0; i ≤ 25; i++)  
{  
    Serial.println(i);  
}
```

int i = 0

Initial value of a variable to control the loop

i ≤ 25

Used to determine the end of the loop

i++

Change of the value after each loop (i++ adds 1)

Use i as the index.

```
int data[5] = {0,1,2,3,4};

void setup() {
  Serial.begin(9600);
}

void loop() {
  for (int i = 0; i < 5; i = i+1)
  {
    Serial.println(data[i]);
    delay(1000);
  }
}
```

i = 0

Initial value of i is 0, the first index

i < 5

Limit i to below 5, the size of the array

data[i]

Prints data[0] the 1st time, data[1] the 2nd time, and so on

What is the value of a?

READ THE CODE

```
int a = 20;
int number[5] = {1, 3, 5, 2, 4};
for (int i = 0; i < 4; i++) {
    a = a - number[i];
}
```

ANSWER

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WHY

i runs 0 to 3, so $a = 20 - 1 - 3 - 5 - 2 = 9$. number[4] is never used.

Lessons 1 to 4, done.

Today we reviewed data types, errors, onscreen, input, if-else and logic, sensors, arrays and for loops.

Thank you!